

Traces of the Other – Are DMT Entities Real?

DMT Phenomenology in the Framework of Conscious Realism

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Abstract.

Encounters with apparently autonomous and intelligent non-human entities are a striking and recurrent feature of high-dose N,N-dimethyltryptamine (DMT) experiences. Such encounters are typically interpreted as complex hallucinations derived from internally generated brain activity, although many features of the DMT state remain a challenge to explain within the standard neuroscientific paradigm. Here, we explore an alternative hypothesis grounded in the conscious realism framework, in which reality consists of interacting conscious agents and perception functions as a species-specific interface rather than a veridical representation of an objective world. Crucially, the space of conscious agents we can normally perceive and interact with represents only a thin slice of the agent network that comprises the totality of reality and is partly determined by representational limits set by the qualia kernel that governs transitions between experiential states.

We propose that DMT induces a profound perturbation of our perceptual interface, expanding the accessible region of experience space and allowing consciousness to reach regions with entirely different dynamical regimes under the qualia kernel. Under such conditions, normally imperceptible agents may influence experience, leaving perceptible “traces” in its dynamics that can be rendered as stable, coherent, and meaningful structure and experienced as the abundantly populated, inordinately complex alternate worlds typical of the DMT state. This framework enables us to distinguish between internally generated hallucinations and experiences exhibiting structured, stable, and causally efficacious dynamics consistent with interactions with normally imperceptible conscious agents. Rather than asserting the reality of DMT entities, we draw on the phenomenology of the DMT state, models of psychedelic brain dynamics, and the formalism of conscious agent theory to derive testable predictions and propose experimental paradigms to assess whether DMT experiences can be constrained by external variables or exhibit non-trivial intersubjective correlations. The outcome of such experiments may have profound implications for our understanding of perception, consciousness, and the structure of reality.

1 Introduction.

For most in the modern West, the idea that we live in a mechanistic and material world built from physical components is accepted as an obvious and undeniable fact. Although materialist ontologies can be traced back to ancient atomism (e.g., Democritus, Epicurus, and Lucretius), it was in early modern Europe that this paradigm became firmly established, as spiritual and mystical descriptions of nature were progressively excluded from scientific discussion and matter planted firmly at the ground of reality. By the nineteenth and twentieth centuries, materialism (or physicalism) had come to dominate Western scientific thought, with physical matter and energy regarded as primary and consciousness typically treated as an emergent phenomenon, in principle reducible to underlying physical, neural, or functional processes [1-3].

But this view certainly isn't universal across all human cultures: If you look to indigenous animistic and shamanistic cultures, Hindu and Buddhist cosmologies, Daoism, and even mystical sects within Islam, Judaism, and Christianity, you won't find an inert, rule-governed super-galactic machine but a living, intentional cosmos permeated by gods, spirits, and other discarnate non-human intelligences [4-7]. Naturally, such magical and mystical beliefs have been roundly rejected by Western materialism, as visions, voices, and other encounters with normally unseen discarnate beings are summarily dismissed as mere hallucination and the symptoms of a mind gone awry.

Yet despite more than two centuries of commitment to this materialist model of reality, the central problem it introduced remains unresolved. While the neuroscientific literature is filled with systematic correlations between conscious experience and brain activity, it has failed to explain how or why subjective conscious experience should arise from objective physical processes [8, 9]. This difficulty, now widely known as the Hard Problem of consciousness, suggests that the assumption of matter's ontological primacy may itself have been a critical misstep in the history of science.

In response, a growing number of contemporary theories reverse this foundational assumption by treating consciousness, rather than matter, as fundamental. Giulio Tononi's Integrated Information Theory (IIT), for example, proposes that consciousness is identical to a system's intrinsic causal structure and, as such, is not emergent from physical substrates so much as realized wherever a system's integrated information is non-zero [10, 11]. More radically, the theory of conscious realism, the framework discussed in this paper, abandons the assumption of mind-independent physical objects altogether, proposing instead that reality is composed entirely of interacting conscious agents [12] . That is, the world consists of (and only of) conscious agents and it is this network of conscious agents that gives rise to the experience of space, time, and the objects within it. The question then is not "How does consciousness emerge from physical matter?" but, rather, "How does the experience of the physical world and its laws, as we humans know them, arise from the interaction of conscious agents?" In this model, the

world we perceive is not fundamental, but more like a user interface that presents a simplified, species-specific model of the underlying conscious agent network [13, 14]. Our perceptions, in this view, are akin to icons on a desktop – they carry information useful for survival, but they do not faithfully depict the true causal structure of reality.

This interface theory of perception is bolstered by evolutionary models showing that natural selection tends to favour fitness over truth. Previous work in evolutionary game theory has demonstrated that perceptual systems tuned purely to objective truth are routinely outcompeted by those tuned to fitness payoffs [14], and theoretical models have shown that, as the complexity of the environment grows, the probability that veridical perceptions are favoured by evolution approaches zero [15, 16]. In short, our perceptions evolved to guide adaptive behaviours, not to reveal truth. As such, it’s entirely plausible that what we take for the “physical world” is merely a species-specific adaptive representation. Space-time itself may be just one data format or “headset” for conscious agents – one that humans happen to use [17]. Indeed, many physicists are now abandoning the assumption that space-time is fundamental. For example, candidate approaches to quantum gravity suggest that space-time breaks down as a valid description of reality at very small scales or high energies, while novel frameworks in fundamental physics increasingly lean towards the possibility that space-time is emergent [18, 19].

If we accept, as a working hypothesis, that consciousness (in the form of interacting conscious agents) is fundamental and that the world we perceive is a mere interface, an obvious question arises: Can we see beyond the interface? Our ordinary perceptions only show us a tiny, filtered slice of reality, populated by those agents that were relevant to our evolutionary ancestors. Conscious realism predicts that there could exist countless other conscious agents – potentially of arbitrarily high complexity or completely unfamiliar form – that we normally do not perceive at all [20]. We might expect that these “hidden” agents do not appear as ordinary objects (or living creatures) in our 3D space-time world because their properties cannot be represented within the constraints of our human sensory interface. The central hypothesis explored in this paper is that certain alterations of consciousness represent perturbations of the interface that enable it to represent structures and dynamics that are informative of the conscious agent network but normally lie beyond its representational reach. In particular, we propose that the powerful naturally-occurring psychedelic N,N-dimethyltryptamine (DMT) may transiently transform the human perceptual interface in such a way that normally “invisible” conscious agents become perceivable.

This hypothesis is motivated by striking phenomenological reports from DMT users: Unlike other classic psychedelics, such as LSD and psilocybin, DMT, at sufficiently high doses, induces a complete transformation of the subjective world: Users commonly describe being transported to what appears to be a fully autonomous alternate reality populated by a panoply of apparently extremely intelligent, non-human entities [21-25].

The standard view in neuroscience is that these experiences are merely particularly elaborate hallucinations. However, as we will discuss, the DMT state possesses a number of characteristics that challenge this assumption. As such, we present this alternative hypothesis as consistent with a consciousness-first ontology, and articulate clear experimental predictions that could falsify (or lend credence to) the notion that some DMT-induced encounters should be treated as perceptions of “other” consciousnesses.

2 The Conscious Realism Model.

Within the conscious realism formalism [17, 20, 26], a conscious agent (CA) has just four assumed properties: qualia, choice, action, and a sequencing of qualia. The possible experiences – qualia – of a CA are represented by a measurable space, $(X, \mathcal{X})$ (a set X together with a sigma-algebra on the set. i.e. a collection of subsets of X satisfying certain closure properties) since, at a minimum, we must speak of the probability that a CA has some experience and a measurable space is a minimal formalism that allows this. Likewise, the possible actions of a CA and the possible states of the world the CA interacts with (including itself) are represented by measurable spaces, $(G, \mathcal{G})$ and $(W, \mathcal{W})$, respectively.

A key intuition is that, within this formalism of conscious agent theory, consciousness involves three processes: perception, decision, and action. In the process of perception, a CA interacts with the world and, as a consequence, has conscious experiences. In the process of decision, a CA chooses what actions to take based on the conscious experiences it has. In the process of action, the CA interacts with the world in light of that decision and affects the state of the world (the network of CAs). These influences are modelled as Markov kernels: If we assume a finite countable set of states, the perception kernel, P , is a (Markov) transition matrix that describes the probability that, for all $x \in X$, a CA transitions to qualia state, x , given a particular state of the world, w . The decision kernel, D , similarly describes the probability that a CA chooses action, g , given a particular qualia state, x . And the action kernel, A , describes the probability that the world (the network of CAs) transitions to state, w , given the action of the CA, g .

So, overall, we can represent a CA by a 7-tuple:

$$CA = ((X, \mathcal{X}), (G, \mathcal{G}), (W, \mathcal{W}), P, D, A, N)$$

where $(X, \mathcal{X})$ is a measurable space of qualia, $(G, \mathcal{G})$ is a measurable space of actions, $(W, \mathcal{W})$ is a measurable space constituting the set of states of the collection of the network of all CAs, N is an integer that counts the sequence of experiences, and

$$P: W \times \mathcal{X} \rightarrow [0,1]$$

$$D: X \times \mathcal{G} \rightarrow [0,1]$$

$$A: G \times \mathcal{W} \rightarrow [0,1]$$

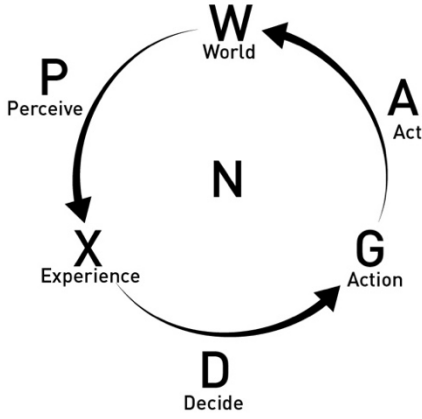
are Markov kernels.

For a conscious agent (CA), we define the qualia kernel Q as the composition of its decision, action, and perception kernels:

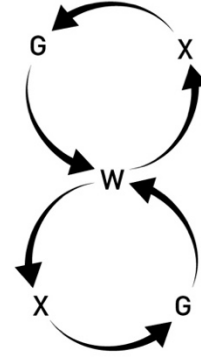
$$Q = DAP: X \times X \rightarrow [0,1].$$

Again assuming a finite set of discrete countable states, $Q(e,e')$ is simply the probability that, given the current state, e , the next state will be e' . In other words, Q is a Markov kernel describing the transition probabilities between experience states. The

triple $(X, \mathcal{X}, Q)$ (i.e. the measurable space of experiences together with the qualia kernel dynamics) thus describes the evolution of the agent's subjective experience.



A conscious agent (CA) has 7 components. The maps P , D , A can be thought of as communication channels.



The world is composed of a network conscious agents which interact by performing actions on the world that can influence the experiences of other agents.

A central concept within this model is that of perception between agents. What does it mean for one agent to perceive another agent? Let agent A have experience space X_A , qualia kernel $Q_A = D_A A_A P_A$, and perception kernel $P_A: W \rightarrow X_A$. Let agent B be defined in the same way, with experience space X_B , qualia kernel $Q_B = D_B A_B P_B$, and action kernel $A_B: G_B \rightarrow W$. Now, we can ask: What does it mean for A to “perceive” B ?

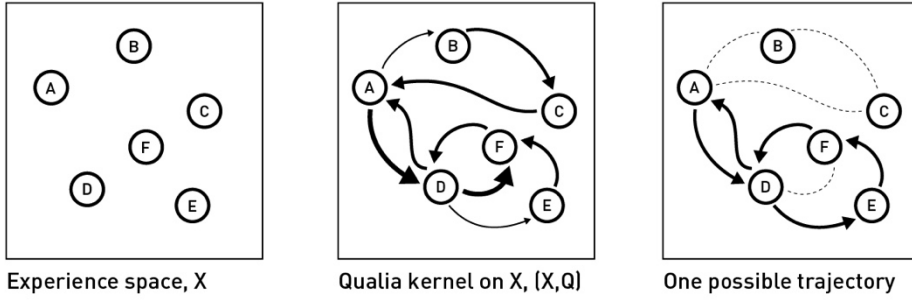
Agent A does not have direct access to the experience space X_B of agent B . Instead, any influence of B on A must be mediated through the world. Specifically, the experiences of B drive its decisions via D_B , which in turn generate actions in G_B . These actions alter the world via A_B , and the resulting world states give rise – through the perception kernel P_A – to probability distributions over states in X_A . Thus, the only channel through which A can “perceive” B is:

$$X_B D_B \rightarrow G_B A_B \rightarrow W P_A \rightarrow X_A$$

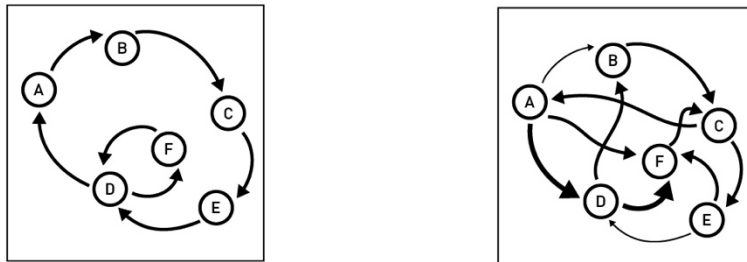
We can use kernel composition to define the effective channel:

$$\Phi := P_A A_B D_B$$

as the induced Markov kernel from X_B to X_A . For each $x_B \in X_B$, $\Phi(x_B, \cdot)$ is a probability measure on X_A that gives the conditional distribution of the next experience for A given that the current experience for B is x_B (i.e. resulting from B's decision and action dynamics followed by A's perception). Thus, Φ specifies how states of B probabilistically influence the experiences of A (if at all), rather than defining a deterministic mapping between their experience spaces. As such, if A's perception kernel P_A is sensitive to some subset of the world states influenced by B's actions, then the dynamics of B can influence the dynamics of A.



The qualia kernel Q equips an experience space X with a transition structure, specifying which experiential states can follow which others. A particular stream of experience corresponds to a trajectory through X governed by these constraints.

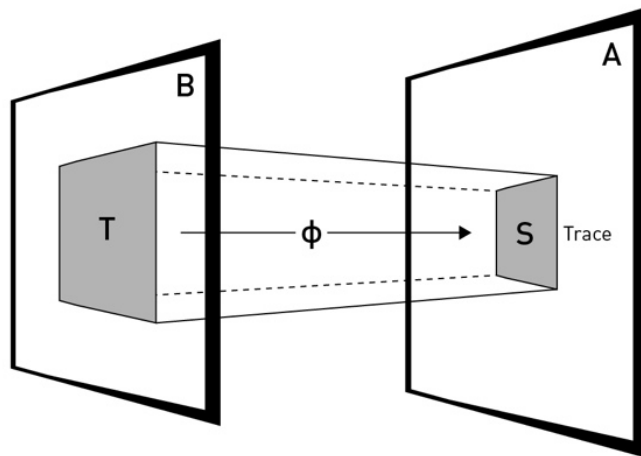


Two conscious agents with different Markovian dynamics. Both agents share the same experience space, but differ in the structure of their qualia kernels. The agent on the left supports relatively sparse and locally organized transitions between experiential states, whereas the agent on the right supports a richer and more interconnected pattern of transitions, allowing a wider range of possible experiential trajectories.

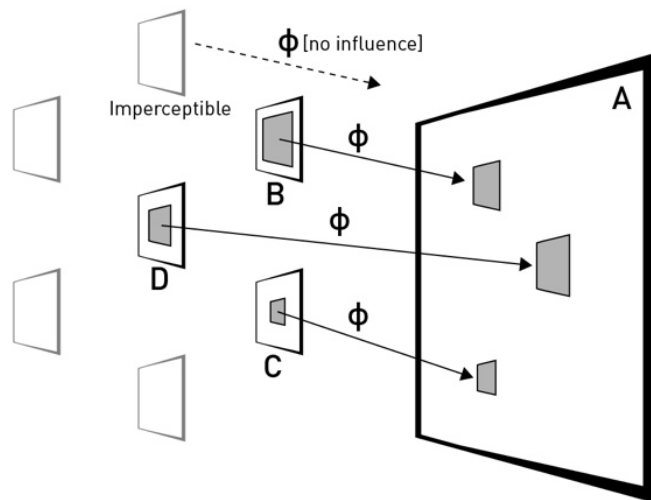
Formally, this relationship can be captured by the notion of a *trace*. Let $S \subseteq X_A$ denote the subset of states in X_A that have a nonzero probability of being the next experience for A under the induced channel $\Phi := P_A A_B D_B$. The dynamics on S , together with their induced transition structure, define a trace kernel on S , obtained by restricting attention to these states and incorporating transitions through states outside S . (S is a trace induced by Q_B and represents the set of states within X_A that are possible under the influence – mediated through the channel Φ – of the actions of B). That is, it describes the effective evolution of those aspects of B’s dynamics that contribute to the dynamics of A’s experience via its perception kernel. This does not imply that A has direct access to B’s internal structure or dynamics, nor even that B’s dynamics are identifiable or representable as such, but simply that A’s experience may exhibit statistical effects that are influenced by B’s behaviour. In short, A perceives B when the actions of B have an impact on the updates of A’s experience.

Crucially, the fact that B performs actions on the world does not in itself imply that those actions will influence the experience of A. P_A is itself constrained: it is only responsive to a subset of world states in W , and if there is no overlap between the world states influenced by B’s actions and those to which P_A is sensitive, then B has no influence on the dynamics of A’s experience. Although B continues to perform actions on the world, none of its effects propagate through P_A to influence A’s experience. As far as A’s experience dynamics are concerned, B plays no role. Or, in short, B is imperceptible to A. More generally, since the perception kernel is sensitive only to a subset of possible world states, by assumption, no agent has access to the full structure of the underlying agent network. Rather, each agent learns or evolves to selectively respond to a restricted range of influences while filtering out the others. In this sense, a conscious agent’s experience may be shaped by only a limited subset of the broader network of agents that comprise reality.

Furthermore, even if there is overlap between the world states affected by B’s actions and those to which P_A is sensitive, the presence of this influence does not guarantee or even imply that A will have any particular type of experience. The possible content of A’s interface is specific to A itself and, although B’s actions on the world might influence A’s experience, the experiences of A and the transitions between them are determined entirely by A’s experience space and qualia kernel. We can’t, for example, make the claim that just because B is a conscious agent that influences A’s experiences, that B will somehow be represented in A’s interface as some kind of “agent” or indeed as any kind of recognisable structure. In fact, in the conscious realism framework, the distinction between “agents” and “non-agents” (e.g. inanimate objects) does not correspond to a fundamental ontological difference (since reality consists of and only of conscious agents). Rather, it reflects differences in how patterns of influence from other conscious agents are organized within an agent’s experience. Since all aspects of the interface are



Conscious agent A perceives conscious agent B when a subset T of B's experience states, via the effective channel ϕ , influence the updates of a subset S (the trace) of A's experience states.



Only conscious agents whose actions have an influence, via ϕ , on the experience space of agent A form some kind of representation within A's interface, whereas others are imperceptible.

generated by the agent’s experience space and qualia kernel, what appears as “agentic” corresponds to patterns of dynamics within the interface that give rise to some qualia of encountering another agent. There is no single minimal, necessary, condition for this. Different constellations of patterns can give rise to such qualia: visual form, motion, voice, intentionality, resistance, responsiveness, felt presence, informational content, and so on. Note that this, necessarily, includes the experience of encountering and interacting with another human being or animal, but also completely different experiences, such as talking to an LLM passing the Turing test, or simply "feeling a presence". And, of course, hallucinations (such as the "internally generated" experience of talking with a deceased relative) are also represented as agentic dynamics in the interface. By contrast, “non-agentic” dynamics correspond to patterns that are more tightly constrained, lower-dimensional, or more directly coupled to the surrounding experience, and which do not exhibit the same degree of apparent independence or internal variability and would include all types of qualia that are not perceived as agents.

Overall, whether the influence of another conscious agent gives rise to agentic or non-agentic dynamics depends not only on that agent’s underlying dynamics, but also on the structure of the observer’s perception channel, experience space, and qualia kernel. As such, the same underlying agent may give rise to agentic dynamics in one observer’s experience and non-agentic or even entirely incoherent dynamics in another’s, reflecting differences in how the trace of its influence is organized and stabilized within each observer’s interface. There is never any direct mapping between the structure and dynamics of an agent and how its trace is represented in the interface of another agent.

3 Evolution of the Interface.

Within the conscious realism framework and under the fitness-beats-truth theorem [14], evolution is understood not as a process that shapes perceptual systems to accurately represent an objective, mind-independent world, but rather as one that tunes the dynamics of conscious agents to maximise fitness. In this view, fitness refers to the capacity of an agent to survive and reproduce within the network of interacting agents, and perceptual experiences are shaped to guide adaptive behaviour rather than to reflect underlying reality [16, 27]. The perception and qualia kernels of an agent are thus sculpted to encode information that is useful for action, not information that is veridical, while simultaneously limiting the region of the agent’s experience space that is typically explored under its dynamics. As a result, an agent’s experiential interface can be understood as a species-specific, fitness-optimised system that both filters and transforms the underlying dynamics of other agents and restricts the range of experiential states that are effectively accessible, thereby shaping a structured domain of experience that supports adaptive interaction with a limited subset of the broader agent network.

In Markovian terms, this corresponds to the dynamics being concentrated within a subset of the experience space that is commonly visited and exhibits stable, recurrent behaviour under the qualia kernel. Under typical conditions, the agent’s experience remains within this region, giving rise to a structured and apparently shared subset of experience that may be understood as a Consensus Reality Space (CRS) across similarly structured agents. If the experience space is imagined as a vast landscape of possible states, the CRS can be viewed as a basin of attraction that both limits which experiences are typically reached and shapes which influences from other agents are able to register within the interface.

Notably, conscious realism suggests that the structure of space-time itself is an emergent property of qualia space dynamics. Recent work by Hoffman et al. has conjectured that the 3+1-dimensional world we experience is not a fundamental feature of objective reality but merely a coordinatization of our interface; the format of experience that evolution has landed upon for tracking and interacting with the kinds of conscious agent dynamics that mattered for survival [20]. Three spatial dimensions might, for example, be just enough to represent the complexity, structure, and dynamical influences of the conscious agents we need to perceive and interact with, without wasting resources on unnecessary complexity and dimensionality. Naturally, this dramatically limits the kinds of conscious agents that can be perceived and rendered in our interface. The obvious question then is, what would it take for a human observer, with a restricted experience space and qualia kernel dynamics, to perceive conscious agent dynamics that are not normally represented within our interface?

An important aspect of the conscious realism model is that it allows us to meaningfully characterize and compare the structure and dynamics of conscious agents. Agents may differ not only in the number of experiential states they can occupy, but also in the pattern of adjacencies and dependencies among those states determined by the qualia kernel. Some agents may have relatively sparse and locally organized adjacency structures, so that changes in experience unfold through limited, sequential transitions. Others may be defined by much denser or more globally constrained adjacency relations, in which many experiential states are mutually adjacent and transitions depend on relations across large portions of the state space at once, rather than on a small number of local features. Crucially, there is no upper bound on the size of the state space of a Markov process, and even for a fixed finite state space there are an uncountably infinite number of distinct transition kernels and, as such, an infinite variety of possible distinct conscious agents. In fact, for a conscious agent with an experience space consisting of n distinct states, the set of all possible Markovian transition structures on those states can be represented as an $n(n - 1)$ -dimensional geometric object known as the Markov polytope M_n [28]. Each point in this polytope corresponds to a distinct Markov kernel, specifying a complete set of probabilistic transition rules between experiential states. Even for modest values of n , the space of possible transition structures is already vast (even for

an experience space with just ten states, the Markov polytope is a 90-dimensional object with ten billion vertices, with each point within the polytope corresponding to a fundamentally different way in which experience could evolve over time). In this sense, the Markov polytope captures an astronomically rich family of possible experiential dynamics, emphasizing that the framework permits an enormous diversity of conscious agents even before any additional structure or constraints are imposed.

As such, reality may consist of a practically infinite number of conscious agents exhibiting an incomprehensibly vast variety of dynamical structures. The overwhelming majority of these would be entirely imperceptible to us, not because of spatial separation or concealment, but because we normally occupy a region of the broader experience space landscape with associated qualia and perception kernel structures that have been sculpted to perceive and represent traces of a thin slice of the conscious agent network. Most conscious agents perform actions on the world that do not give rise to states to which our perception channel is sensitive. And even when such influence is present, it need not give rise to coherent or intelligible patterns within human experience. If the dynamics induced by such agents are not well supported by the structure of our experience space and qualia kernel (for example, by requiring richer dependencies, higher-dimensional relations, or alternative transition structure than our interface can support), then their influence may fail to appear as stable or interpretable patterns, and may instead be filtered out, transformed, or rendered as noise or other non-agentic dynamics. In this sense, there likely exist a vast number of conscious agents whose influence either does not register in our experience at all, or whose actions induce dynamics within our experience space that exceed the range of patterns that our interface can sustain as coherent structure. Importantly, however, this should not be understood as implying that there exists a more complete or “truer” form of experience prior to this filtering: the world state itself is not directly accessible, and there is no canonical or “correct” way of representing it. Rather, the perceptual interface both constrains and constructs experience, selectively highlighting, organizing, and in some cases enriching aspects of the underlying dynamics of the conscious agent network in forms suited to the agent.

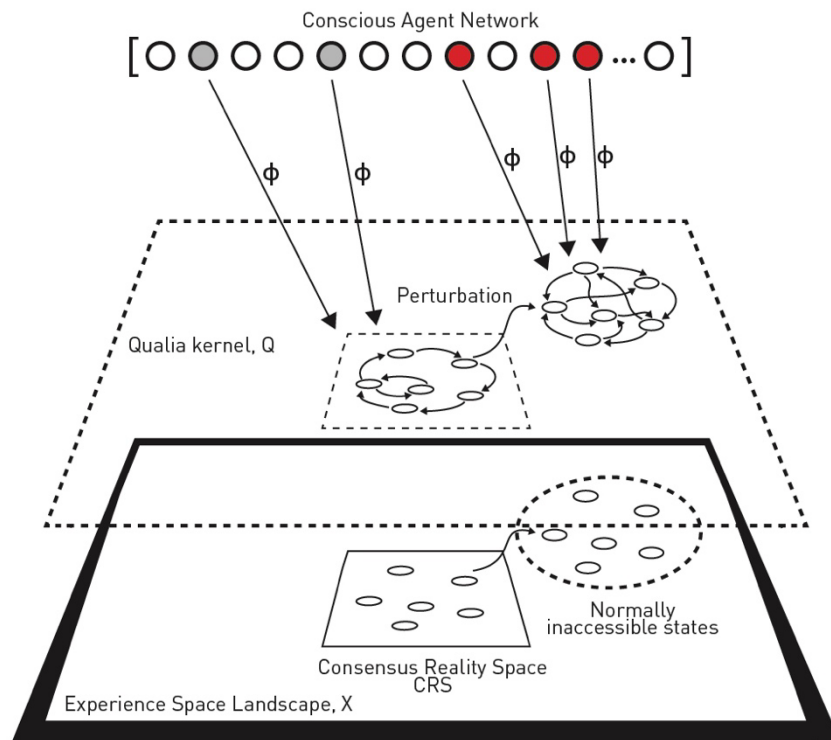
However, there is no reason to assume that the perceptual and dynamical constraints imposed by our qualia and perception kernels within the CRS should apply outside of it. Whereas the applied qualia and perception kernels shaped by evolutionary pressures determine the range and types of conscious agents we can perceive within the CRS, absent such pressures within the broader experience space the range of perceivable agents might be entirely different and may give rise to patterns of influence that do not simply extend those found in consensus reality, but instead reflect entirely different regimes of dynamical structure. As such, perceiving normally imperceptible conscious agents might be as simple as applying a robust perturbation of our experience space to reach out of the CRS and open up states and dynamics such that a normally imperceptible agent’s

influence on our experience space can be rendered as stable, coherent, and meaningful structure that would normally be beyond our representational reach.

To use an analogy: If you open a JPEG file on your computer using a photo app, you will be presented with an image that immediately makes sense. If you open the same file using a word processor app, you’re likely to be greeted by an entirely incomprehensible mess of letters, numbers, and other exotic symbols. Both apps are using the same data encoded in the file but how they represent this data is entirely different. The photo app doesn’t generate a more accurate or true representation of the file data, but it does render the data in a form that is obviously more meaningful to a human observer. In an analogous manner, while we can never have access to the internal dynamics of another agent, the ability of our interface to represent the “data” received via our perception channel in a meaningful form is determined by the structure of our experience space, qualia kernel dynamics, and our perception kernel. For example, a data structure that could usefully be represented as a 5-dimensional object cannot be rendered as such in the human interface and might be ignored or treated as noise. In order to form a coherent and meaningful representation of such data, the human mind might need to enter an extraordinarily unusual state in which, for example, its standard 3+1-dimensional perceptual interface is replaced by a higher-dimensional or non-Euclidean format. During such an altered state, the forms and content comprising the person’s experience would radically transform, reflecting the structure of the previously hidden agent’s influence on the experience space of the observer.

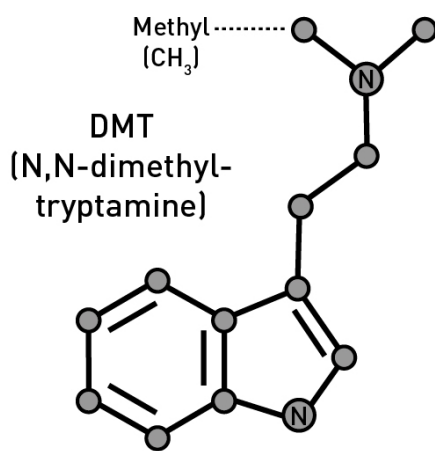
4 Traces of the Other.

There are many examples of systems in which a sufficiently robust perturbation (such as an increase in an energy parameter) causes an effective expansion of the accessible state space, as new states or degrees of freedom become dynamically accessible: liquid–gas phase transitions [29, 30]; activation of rotational and vibrational modes in molecules [31]; the activation of higher-energy degrees of freedom in particle physics [32]; and the flattening of the energy landscape with increasing ‘temperature’ in recurrent neural networks [33, 34]. Likewise, multiple human studies have revealed that psychedelics, by stimulating a subset of neurons in the deeper layers of the cortex, dramatically increase the entropy of cortical activity and induce a more fluid “hot” state in which the cortex explores an increased repertoire of states and generally exhibits higher levels of neural complexity [35-42]. The subjective effects of psychedelics comport well with the observed effects on neural activity, indicating that the experience space is nudged into a more fluid and dynamically exploratory state outside of the CRS that at least hints at the kinds of dynamic complexity that might be required to support traces of normally imperceptible conscious agents. However, at usual dose levels at least, most psychedelic users rarely report encounters that might be construed as the perception of otherwise undetectable conscious agents. It’s plausible that a more profound restructuring of the



The qualia kernel, Q (composed of the perception, decision, and action kernels) imposes the dynamical structure on the states of the experience space, effectively restricting the conscious agent to a limited region (CRS) of the broader experience space landscape, X. However, the dynamics imposed by Q within the CRS do not apply outside of it. As such, a robust perturbation of the conscious agent may drive it into areas of X with entirely different dynamical regimes where the qualia kernel dynamics can represent traces of normally imperceptible conscious agents. (Grey circles represent CAs perceivable from within the CRS; red circles represent CAs perceivable in the post-perturbation state; non-coloured circles are imperceptible CAs)

experience space might be required to allow it to support traces of dynamics that are exotic when compared to those normally encountered in consensus reality, and perhaps higher-dimensional geometric and topological structures of conscious agents with unusual Markovian dynamics. Naturally, such a profound perturbation of our experience space wouldn't simply cause otherwise hidden conscious agents to "pop up" as supplementary icons in our interface, but would likely entail an almost complete restructuring of the content, dynamics, and flow of conscious experience itself – a complete replacement of the normal interface with one that transcends the usual topological and geometric structures imposed by our standard 3+1D interface. One such candidate molecule that *does* appear to induce such a perturbation of our experience space is the naturally-occurring psychedelic, N,N-dimethyltryptamine, DMT.



DMT is set apart from the other classical psychedelics, such as LSD and psilocybin, in that, when vaporised or injected, it rapidly and reliably induces a state in which users report being completely immersed in a bizarre, often bewildering “alternate reality” that feels just as real as (and often more real than) ordinary waking life [21-24, 43]. Users commonly describe entering otherworldly landscapes replete with inordinately complex, fractal visuals, and impossible higher-dimensional geometries. And, notably, these alternate worlds are often populated

with a variety of apparently highly intelligent non-human entities. A large-scale analysis of 3,778 DMT trip reports found that entity encounters were reported in about 45% of experiences, often involving beings described as humanoid, alien, or spiritual in nature (commonly described as “machine elves,” insectoid or reptilian creatures, deities, jesters, etc.) [44]. Other studies report similar entity prevalences [22], although one naturalistic setting study reported non-human entity encounters in 72% of subjects [45]. Many of these entities are reported to behave autonomously and to communicate with the experiencer (through language or telepathy), sometimes imparting information or lessons. Remarkably, across thousands of reports, certain motifs repeat: entering a tunnel or portal; arriving in a “waiting room” or hyper-technological chamber; meeting a guiding or “pedagogical” being; experiencing a sense of higher-dimensional space; and witnessing scenes of advanced technology or alien architecture.

The standard explanation for these experiences is that they’re a particularly elaborate form of hallucination or perhaps related to dream visions [46]. Evolutionary neurobiology suggests that the human brain possesses innate cognitive modules for detecting agency, faces, and social presence [47, 48]. It has been hypothesised that these modules might start generating imagery of autonomous beings when top-down control of perception is

loosened (as in psychedelic states) [49]. Indeed, Timmermann and others have hypothesized that DMT suppresses high-level priors and the default mode network in the brain, leading to a flood of unconstrained bottom-up signalling (especially from the medial temporal lobe, where memory, emotion, and face/object recognition reside) [50, 51]. This could produce fully realized “entity” experiences by essentially allowing the brain’s pattern detectors to impute agency and form to random neural noise.

However, similar loosening of top-down control occurs during dream sleep and in certain psychotic conditions [52-54]. In both states, neuroimaging studies implicate medial temporal and higher-order visual networks as key contributors to internally generated visual imagery [53-55]. Dreams, in particular, tend to be continuations of the normal waking world, with the brain drawing on learned and stored object models when rendering dream environments, which are generally populated by humans and animals encountered in everyday life [56-59]. Likewise, the hallucinations of psychotic patients tend to be “similar to real visual perceptions, as most of them are normal sized, solid, and anchored in external space,” and, as in dreams, the majority of hallucinated characters are “normal-sized, normal-appearing people.”[60, 61] This pattern is unsurprising since, within the materialist framework at least, the brain’s innate cognitive modules evolved to detect and model other humans and animals encountered in the ancestral environment, and human neurodevelopment involves learning to construct a subjective world as a model of that environment and its inhabitants. Consistent with this, evidence from dream research in children suggests that the brain’s ability to construct a coherent subjective world and its occupants in the absence of sensory input is not fully innate, but develops gradually from infancy through adolescence [62, 63]. Accordingly, when freed from top-down constraints, it is entirely expected that spontaneously generated visual imagery would predominantly consist of humans and other familiar animals.

Strikingly, however, humans – by far the most commonly encountered organism in the natural waking environment – are only rarely encountered in the DMT state (<5%) [44, 45, 64], which tend to be populated by beings that are not only neither human nor animal, but far transcend the human in their complexity and behaviour. It remains a challenge to explain how and why, when perturbed by DMT, the human brain should so consistently generate imagery of apparently intelligent beings that often have absolutely no obvious referent in the human environment, inhabiting entirely alien worlds of staggering structural, dynamical, and narrative complexity that bear no relationship whatsoever to the normal waking world or to human experience, either now or at any time in our evolutionary history. So, whilst the standard hallucination explanation must remain as the default, most parsimonious explanation for DMT’s effects, considering and empirically testing alternative hypotheses concerning the underlying structure and origin of these experiences is both warranted and scientifically tractable.

Our working hypothesis is that some DMT entities might correspond to traces of conscious agents with Markovian dynamics that are, compared to consensus reality, highly unusual or exotic and, as such, are normally imperceptible but which can be perceived when DMT alters the structure and dynamics of our interface, allowing human consciousness to reach areas of the experience space landscape outside of the CRS where some aspects of the evolved constraints imposed by our qualia and perception kernels no longer apply. We hypothesise that, in these states, the qualia kernel may become capable of representing traces of normally imperceptible conscious agent dynamics as stable, coherent, meaningful, and agentic forms. Indeed, psychedelics, including DMT, appear to induce a flattening of the cortical state space landscape, reducing the energetic barrier for transition between cortical states and allowing the cortex to reach and explore novel states not available during normal waking consciousness [41, 65]. Although we cannot assume a direct mapping between cortical dynamics observed in neuroimaging data and the effects on conscious agent dynamics, such data does represent, in some aspect at least, the way that some of the effects of DMT are represented in the interface and comports with the idea that, under the influence of DMT, consciousness is able to reach regions of the experience space landscape where the perception of otherwise imperceptible conscious agents becomes possible. And, indeed, certain patterns in DMT reports align with what one might predict when a human-level conscious agent observes and interacts with conscious agents that live in and manipulate an entirely different experiential manifold than we do:

4.1 Higher-Dimensional Phenomenology.

Many DMT users report explicit experiences of extra spatial dimensions or non-Euclidean geometry, describing the space as “more than 3D” or “hyperdimensional.” For example, one report recounts, “There were way, way more directions in space to look at than there should have been... More than three dimensions”[66]. Another user reports, “They [the entities] occupied the fourth spatial dimension, yet I could only perceive them through my understanding of three-dimensional space... as if I was viewing a tesseract rotating in 4D.[67]” These descriptions of a perceptual space expanding or warping in dimensionality are exactly the kind of effect we’d expect if the mind is interfacing with dynamics that don’t fit into a Euclidean 3D field.

Andrés Gómez-Emilsson of the Qualia Research Institute (QRI), an independent think tank devoted to studying and formally describing the structure and quality of conscious experience, has proposed a complementary account in which DMT drives consciousness into a regime of extreme energetic excitation, allowing phenomenal space to reorganize in ways that are not available in ordinary experience [68, 69]. In this model, a moment of experience can be understood as a distribution of a finite “energy” across perceptual and cognitive features, which is continuously redistributed over time. Under ordinary conditions, phenomenal space reflects a balance between the rate at which the

mind detects symmetries and relational structure, and the rate at which such structure decays. Psychedelics, and DMT in particular, are proposed to significantly slow this decay while increasing the sensitivity to relational structure, allowing patterns to accumulate faster than they can be dissipated. As a result, the internal dynamics of experience begin to exceed what can be stably represented within a three-dimensional geometry, and phenomenal space reorganizes into forms characterized by non-Euclidean curvature, higher-dimensional symmetries, and increasingly complex relational structure. The result is a regime in which experiential objects can exhibit coherence and stability despite possessing relational properties that cannot be faithfully represented within three dimensions. Interpreted through the conscious realism framework, this suggests that the DMT user’s consciousness has been pushed into a region of experience space outside of the CRS that supports representations and transition dynamics that are normally inaccessible in human experience [70, 71].

4.2 Extreme Complexity and “Otherness”.

If DMT allows us to perceive conscious agents with entirely different Markovian dynamics to our own consciousness, we might expect them to appear (compared to normal human or animal behaviour) bewilderingly complex and entirely alien. The subjective impression might be as if witnessing a supremely advanced technology or intelligence, far beyond human capabilities. Indeed, many entity encounters on DMT involve beings who manufacture or manipulate impossibly complex objects in front of the user, as if to demonstrate their abilities. One such report describes an entity “materializing the most complex and impossible shapes...transferring as much mathematical and geometric knowledge as fast as possible... I was getting the feeling that this information was basic knowledge in their world, like us teaching our toddlers squares, circles, and triangles.” Another account speaks of “a liquid mind ecology of staggering and alien complexity [with] geometries transcending what is even vaguely feasible in this three-dimensional mundane world... Crowding and cramming themselves into my field of vision were thousands upon thousands of beings of every imaginable sort and many that were completely unimaginable... The primordial intelligence being manifest before me was palpable, undeniable, transcendently amazing... It was like being entertained by the 76,000-piece orchestra of an alien civilization in whose classical music each note is not merely a musical tone, but an entire world, each just as intricate and nuanced as our own.”[72]

These descriptions align with the idea that the operations performed by these entities appear miraculous only because we observe them from our low-dimensional perspective. Analogously, a Flatlander seeing a 3D being perform a simple rotation might be baffled by the shape’s apparent changes. In our case, operations that are trivial for an agent in a higher-dimensional or higher-complexity experiential space might look to us like god-like feats. For example, the amplituhedron is a geometric object defined in a high-dimensional abstract space entirely outside ordinary spacetime, yet it encodes the full

complexity of particle scattering processes that appear extraordinarily complicated when computed using traditional spacetime-based methods [73]. From our low-dimensional perspective, the scattering process seems exceedingly complex; from the higher-dimensional perspective, it's elegant and simple. Analogously, DMT entities might not be “super-intelligent” in a human IQ sense; they might simply be operating in a higher-dimensional experiential space, performing operations that are trivial from their own perspective but exquisitely complex from ours.

Our model specifically does not assume that these agents possess extraordinary human-like intellect but that they may have dynamical skills (such as manipulating complex multi-dimensional visual forms) that transcend those of any human. For example, Gómez-Emilsson reports the rapid and error-free colouring of extremely complex, dynamically changing surfaces such that no adjacent regions share the same colour – a process closely related to the four-colour problem, but applied to highly complex, non-Euclidean, and often hyperbolic surfaces [74]. For humans, solving such problems is a slow and cognitively demanding process, becoming intractable as the surface grows or departs from Euclidean geometry. However, in the DMT state, Gómez-Emilsson describes entities colouring surfaces almost instantaneously and repeatedly, as if implementing a form of parallel, constraint-satisfying computation far exceeding the capabilities of even the brightest of humans, in exquisite command of the exotic geometric and topological structure of the space they occupy.

Another frequently reported “trick” of DMT entities is the ability to simultaneously change every aspect of their form (colour, shape, movement, etc.) in a coordinated manner without losing coherence. Human mental imagery struggles to keep track of even a single aspect of a single object (we normally serialize changes). But entities are reported to, for example, split into dozens of smaller entities and then recombine, or undergo rapid metamorphosis while still clearly being the same entity. This suggests a global coherence to their form that might correspond to an underlying structure with many degrees of freedom orchestrated together – again consistent with a conscious agent with highly unusual Markovian dynamics.

4.3 Independent Agency and Interaction.

If DMT does indeed allow us to perceive and interact with normally imperceptible conscious agents, we would expect the entities to exhibit behaviours or provide information consistent with such an interaction. DMT users regularly describe entities that *seem* entirely autonomous, in that their behaviour is impossible to predict or anticipate, endlessly surprising, and entirely outside of the user's control. Many users report being “taught” by entities, which often takes the form of bewilderingly complex visual performances (such as manifesting complex high-dimensional objects) or “downloads” of vast amounts of highly complex information [75]. There are also anecdotal accounts of multiple individuals encountering the same entity in separate DMT sessions. If such shared

encounters are genuine traces of the same conscious agent(s), then there should indeed be correlated aspects of their form and behaviour. Of course, anecdotes are just that, but such reports will motivate our proposed experimental studies later.

It is important to emphasize that the connections drawn above between DMT phenomenology and the conscious realism framework are suggestive rather than conclusive. Our task, therefore, is to develop experimental approaches that can distinguish between different classes of experience, in particular between those that exhibit the characteristics of interaction with other agents and those that do not. In standard neuroscientific terms, a hallucination is typically defined as a perception that is not appropriately coupled to sensory inputs. More broadly, all perceptions are constructed by the brain, which does not have direct access to an external world, but veridical perceptions are those that are adaptively connected to environmental structure, whereas hallucinations are not. Within conscious realism, *all* perceptions arise from interactions with other conscious agents, and this includes what are ordinarily termed hallucinations. However, this does not preclude the use of empirical methods to distinguish between different types of experiential dynamics within the interface.

Recall that an agent A is said to perceive agent B when the actions of B , mediated through the channel $\Phi := P_A A_B D_B$, influence the updates of A 's experience. In such cases, a subset $S \subseteq X_A$ of A 's experience space supports a trace of B 's dynamics, and the resulting transition structure on S reflects the influence of B 's behaviour. As such, if an entity encounter in the DMT state corresponds to a trace of a conscious agent that is normally imperceptible (as hypothesised), we should be able to detect evidence of this influence, which might include some subset of the following: an entity's ability to introduce novel or unanticipated structure into experience; persistence or coherence over time; behaviour that is not trivially reducible to the subject's prior expectations or intentions; and the capacity to influence other elements of the experiential interface, potentially including other agents. In addition, such encounters may exhibit some degree of intersubjective correspondence across observers. Note that this definition of a "real" entity encounter mirrors how we might ordinarily distinguish veridical perception from hallucination within standard physicalist frameworks. In both cases, the key criteria for establishing veridicality are the presence of structured, stable, and causally efficacious dynamics that are not reducible to the subject's internal processes, and which exhibit some degree of independence and intersubjective consistency. The difference, therefore, lies not in the criteria used to identify such experiences, nor in the experimental paradigms required to study them, but in the theoretical interpretation of what these features signify. In the CAT framework, they are understood in terms of traces of other agents' dynamics within the subject's experience, rather than in terms of correspondence with an objective physical world.

The next section proposes several experimental approaches, inspired by the conscious agent theory (CAT) framework, to test the nature of DMT entities. This kind of research has recently been rendered more feasible by the development of techniques that can extend the DMT trip from a few minutes to an hour or longer. The extended-state DMT (DMTx) technique, first proposed by Gallimore and Strassman [76], employs target-controlled intravenous infusion to establish and maintain a stable brain DMT concentration over time. The technique has already been shown to be safe, effective, and tolerable, with stable psychedelic effects maintained for up to 120 minutes in a research setting [77-79]. This extended format gives the subject time to explore, observe, and even conduct tasks in the DMT state in a way that the standard trip doesn't allow. With the ability to pause and resume the experience by modulating the infusion rate, DMTx offers a degree of experimental control unprecedented in psychedelic research. The standard approach to studying the phenomenology of DMT hasn't really changed from the earliest studies in the mid-20th century to the present day: Administer the drug to a volunteer (often selected only on the basis of them having had some prior experience with psychedelics) and then, immediately following their experience, prompt them to report their experiences using a combination of questionnaires and free verbal or written accounts [23]. We ought, instead, to be sending mathematicians to analyse the apparent non-Euclidean structure, symmetry groups, and invariant features of the space; linguists and semioticians to study the structured language, symbolic systems, and apparent grammars used by the entities and embedded in the environment; anthropologists and comparative mythologists to identify recurring motifs, archetypal forms, and cross-cultural regularities; computer scientists and complexity theorists, physicists, phenomenologists and philosophers of mind. The DMTx technique will also be indispensable in conducting experiments aimed at rigorously testing the hypothesis that the DMT space and its occupants aren't merely scientifically interesting, but that this molecule allows our perceptual systems to support and render traces of normally imperceptible conscious agents.

An important aspect of the conscious realism formalism is that the experience of any conscious agent at a given point in time is represented as a point in a space of possible experiences. Each point in this space represents one full moment of experience for the agent in question. As an example, we assume that, for a human being experiencing something like waking consensus reality with their eyes open, a moment of experience will include their current visual experience, inferred 3D object representations, some audio qualia, some conceptual interpretation of experience (potentially represented as words and images), emotional content (likely including both primitives such as physical tensions, as well as some interpretive layers and reactions), mental imagery, and so on. In a given parametrization, this can, for example, be thought of as a very long string of numbers encoding the current configuration of qualia in the repertoire of the agent. As such, an exhaustive representation of a single moment of experience, let alone a path through experience space, is a complex object defined by a very large number of

parameters. We therefore expect that it will rarely, if ever, be useful to work with anything like an exhaustive or fully general state space.

Instead, for any given experiment utilizing the Conscious Realism framework under CAT, part of what goes into defining the experimental protocol is the construction of a tractable state-space representation capable of capturing the phenomena of interest and producing measurable observables. These might include, for example, specific properties of statements that the subject makes about their experience; overlaps between descriptions of experience across subjects; numerical responses, or, in some cases, perhaps even neuroimaging results. In other words, the formalism is not applied by attempting to model the totality of experience in full detail, but by selecting an experimentally relevant representation of experience that is rich enough to capture the structure of the phenomenon of interest while still allowing mathematically precise analysis and empirical comparison.

5 Proposed Experimental Research Programs.

Rather than a single decisive experiment, we suggest a number of complementary experimental paradigms designed to test whether DMT experiences exhibit signatures that are difficult to reconcile with entirely hallucinatory perceptions. These naturally fall into two classes: experiments involving a single subject, and experiments involving multiple subjects.

5.1 Single-subject experiments.

Single-subject paradigms are designed to test whether features of a DMT experience are systematically constrained by variables outside the subject’s own cognitive and perceptual processes (and, as such, may represent the influence of normally imperceptible conscious agents). These experiments are designed to be conducted in well-controlled settings using the extended-state DMT infusion technique (DMTx) to allow sufficient interaction time.

One class of experiments examines whether an entity encountered during a DMT experience can reliably track or respond to an external variable unknown to the subject. The variable may be randomly generated, time-varying, and blinded to both subject and experimenter during the session. The key question is whether aspects of the experience (visual features, behaviours attributed to an entity, or reported changes over time) show statistically significant correspondence with the hidden variable beyond what would be expected by chance. For example, since the DMT state is a characteristically visual and intensely colourful experience, a pair of colours might be chosen as a binary variable. A computer, isolated in a separate room from both the subject and experimenters, randomly switches between two states (a blue and a yellow screen, for example). If an entity is external and cooperative, it may be able to express a stable blue/yellow modulation

that tracks the hidden variable. Standard controls such as randomization, blinding, pre-registration of analysis criteria, and comparison with control conditions (e.g., imagination or other psychedelic states) are also essential.

A major limitation of this kind of experiment is that it requires the active cooperation of the entity, which is obviously not guaranteed. This is where selective recruiting would be invaluable: There are many anecdotal reports of individuals who claim to regularly interact with the same entity, even to the point of developing a relationship with the entity over many months or longer. Such subjects would be ideal candidates for this kind of experiment without compromising its scientific validity. The experiment also relies on the assumption that the entity would have access to the external variable. However, this limitation can be obviated using another subject (discussed in the following section).

A second class of single-subject experiments focuses not on information transfer, but on the structure of the experience itself. Since entities encountered in the DMT state are often described as exhibiting remarkable coherence, agency, or apparent intelligence, it's also natural to ask whether they might display abilities that exceed those of humans. For example, Rodriguez [80] suggested that DMT entities could be asked to perform mathematical operations (such as finding the prime factors of large integers) that humans cannot carry out mentally. However, such symbolic and sequential reasoning tasks primarily test how effectively a human subject can use their own cognitive machinery. Our model does not predict that DMT entities are more intelligent in the familiar human sense, but rather that they may instantiate dynamical processes beyond those normally accessible to human consciousness. As such, the relevant question is not "Are you smart?", but "What kinds of dynamical structures can you stably instantiate and control?" From this perspective, the appropriate tests are not difficult symbolic problems, but probes of an entity's ability to instantiate, maintain, and transform globally constrained, high-dimensional visual or geometric structures whose coherence requires transition dynamics far richer than those supported by ordinary human cognition. Such tests might include:

Global constraint satisfaction: Can an entity colour a complex surface so that no adjacent regions share the same colour (as described by Gomez-Emilsson)?

Stability across simultaneous transformations: Can a DMT entity remain coherent and identifiable while undergoing multiple independent visual changes (such as shape, colour, motion, or topology) at the same time, rather than handling them sequentially or collapsing into incoherence?

Long-range temporal coherence: Can a DMT entity initiate a structured visual process (such as constructing a complex geometric structure), allow it to be interrupted and displaced by unrelated experiences, and later resume or complete it without any explicit reminder or cue from the subject?

Obviously, rather than simply recruiting DMT users with substantial experience interacting with entities, these kinds of tests would certainly require individuals who also possess substantial mathematical knowledge and experience. And, of course, we'd be relying on the subjective reports of the subjects to determine whether or not a particular task has been completed successfully by the entity (this highlights why we're proposing a number of complementary experimental paradigms rather than reaching for a single decisive experiment).

5.2 Multiple-subject experiments.

Whereas single-subject experiments test for external constraint on an individual experience, multi-subject paradigms test for shared structure across experiences. In these experiments, multiple subjects (isolated from one another) undergo DMT sessions under identical conditions. Their reports are later analyzed for correlations, shared features, or structured correspondences beyond what would be expected by chance. The aim is not to establish identical experiences, but to test whether independent subjects exhibit non-trivial convergence in reported entities, environments, or informational content. Again, selective recruiting would likely improve the chances of success: Couples who claim to have shared experiences, particularly if it involves interacting with the same entity, might be recruited to test for correlations between individual experiences. Such experimental set ups might include information deposit and retrieval, in which one of the subjects deposits information into the DMT space, such as by communicating a randomly generated word or multiple-digit integer to an entity, which is subsequently retrieved from the entity by the other subject.

As with single-subject paradigms, methodological safeguards such as blinding, standardized reporting frameworks, and preregistered analysis criteria are essential. Across all paradigms, careful attention must be paid to expectation effects, experimenter bias, and the interpretive flexibility of retrospective reports. Null results are informative, and positive results would require cautious interpretation. The goal of these proposed programs is not to presuppose any particular metaphysical conclusion, but to clarify which features of DMT experiences can be systematically constrained, correlated, or reproduced, and which cannot.

6 Conclusion.

It is a direct consequence of the conscious realism formalism that the space of possible conscious agents likely vastly exceeds the subset accessible to ordinary human perception. Since perception, within this framework, is an interface shaped by evolutionary pressures rather than a transparent window onto objective reality, there is no reason to assume that the agents and dynamics most relevant to human survival exhaust the space of what exists. The question, then, is not whether such agents could exist in principle, but under what conditions traces of their dynamics might become accessible to human experience.

The DMT state represents an unusually robust and reproducible perturbation of the human perceptual interface that reliably produces experiences of exquisitely complex, coherently structured environments, and apparently autonomous entities. While the standard hallucination explanation remains most widely held, it struggles to explain how and why the human brain, when perturbed by DMT, so reliably induces such phenomenology that is so complex and coherent and yet, in many ways, entirely disjoint from normal human experience. Conscious realism provides an alternative interpretive framework in which such experiences could, in principle, be understood as traces of other conscious agents whose internal dynamics exceed those ordinarily representable within human experience. Importantly, this interpretation does not depend on attributing human-like intelligence to DMT entities, but instead on differences in the structure and complexity of their underlying experiential dynamics, which can be articulated formally using the language of Markovian dynamics. By framing “alienness” in terms of adjacency structure, transition dependencies, and representational limits, we can set aside culturally specific and often poorly defined concepts, such as “plant spirits” or “trans-dimensional beings”, and instead ground the discussion in a precise mathematical framework. As such, the central question is no longer simply whether DMT entities are “real”, but whether the human perceptual interface is capable, even temporarily, of perceiving normally imperceptible conscious agents and hosting traces of conscious-agent dynamics that normally lie beyond its representational reach.

Crucially, the perspective developed here is not insulated from experimental scrutiny. The possibility that DMT experiences might be systematically constrained by variables external to the subject, or might exhibit nontrivial structure across individuals, leads naturally to experimental paradigms that are both falsifiable and methodologically sound. Negative results would strongly constrain external-agent interpretations, while positive results could have far-reaching implications for our understanding of perception, consciousness, and, perhaps even, reality itself. With the recent development and validation of the extended-state DMT infusion technique, DMTx, such investigations are now feasible.

Ultimately, our purpose here is not to claim that DMT entities represent external conscious agents, but to show that, given a formal theory in which consciousness is fundamental, such a possibility can be coherently formulated and experimentally investigated. Whether this line of inquiry ultimately reinforces the standard hallucination model or forces us to consider a more radical revision of our assumptions remains an open question. What matters, from a scientific perspective, is that the question can be asked precisely, tested rigorously, and approached with both openness and restraint.

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